# Gaussian Processes

*Bayesian Modeling with Distributions over Functions*

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# The Shortcomings of Regression

How can we do Bayesian inference when we are unsure of the functional form?

*Ordinary regression assumes the response is linear in the features.*

**BAYESIAN LINEAR REGRESSION**

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,\ \varepsilon_i \sim N(0, \sigma^2),\ \beta \sim p(\beta)$$

## WHERE THIS FALLS SHORT

### Functional form chosen in advance

Must commit to linear, quadratic, cubic, etc before seeing the data.

### Misspecification risk

If the truth isn't in the chosen class, fit and inference degrade.

### Limited flexibility

A finite coefficient vector may not capture complex nonlinear structure.

### Uncertainty on the wrong object

We get uncertainty on β, but our real unknown is the function itself.

# Why Gaussian Processes?

What if we want a prior over functions, not parameters?

### Parametric approach

Assume a fixed form, e.g.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

- Choose degree / basis upfront
- Model misspecification if wrong
- Hard to express “smooth but unknown”

### Gaussian Process approach

Place a prior directly on *f*:

$$y = f(x) + \varepsilon \quad where \quad f \sim GP(m, k)$$

- Nonparametric: no fixed basis
- Encodes smoothness via kernel
- Gives structured uncertainty while remaining very flexible

# What Is a Gaussian Process?

**DEFINITION**

A stochastic process $\{f(x) : x \in \mathcal{X}\}$ is a **Gaussian process** if, for every finite collection of inputs $x_1, \ldots, x_n \in \mathcal{X}$, the vector $(f(x_1), \ldots, f(x_n))$ has a multivariate normal distribution. The multivariate normal is governed by a **mean function** $m(x)$ and a covariance matrix that comes from the **kernel function** $k(x, x')$.

$$f \sim \mathcal{GP}\,(\,m(\cdot),\ k(\cdot, \cdot)\,)$$

**At any finite set of inputs**

$$(f(x_1), \ldots, f(x_n)) \sim \mathcal{N}(m, K)$$

with $(m)_i = m(x_i)$ and $(K)_{ij} = k(x_i, x_j)$.

**Two ingredients**

- $m(x) = \mathbb{E}[f(x)]$ — prior average behavior
- $k(x, x') = \mathrm{Cov}(f(x), f(x'))$ encodes smoothness & scale

*The conceptual shift: from a prior on coefficients to a prior on functions.*

# The RBF (Squared Exponential) Kernel

*The kernel determines how strongly two function values are related under the prior.*

$$k(x, x') = \alpha^2 \exp\left(-\|x - x'\|^2 / 2\ell^2\right)$$

**α² — amplitude**

$k(x, x) = \alpha^2$ — marginal variance

Typical vertical size of function fluctuations.

**ℓ — lengthscale**

$\ell$ — rate of correlation decay

How far the influence of one point reaches.

**IF x AND x′ ARE CLOSE**

$k(x, x') \approx \alpha^2$ — strong correlation

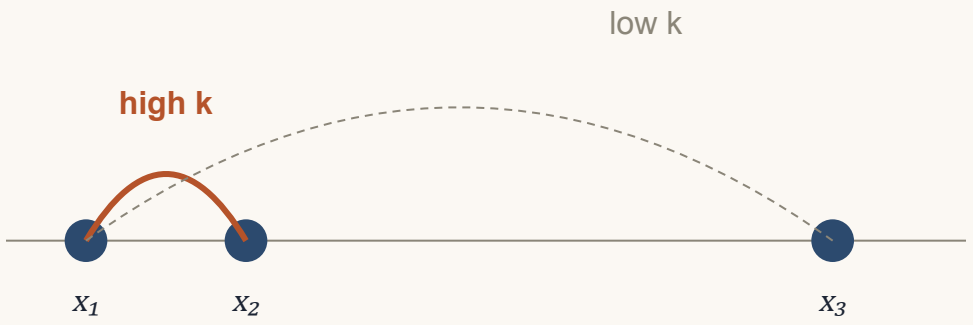
*Nearby inputs → similar function values.*

**IF x AND x′ ARE FAR APART**

$k(x, x') \approx 0$ — nearly independent

*Distant inputs provide little info about each other.*

**HOW IT SHAPES THE PRIOR**



# Bayesian GP Regression

**PRIOR ON THE FUNCTION**

$$f \sim \mathcal{GP}(m, k_\theta)$$

θ = (α, ℓ) are the kernel hyperparameters.

**LIKELIHOOD**

$$y_i \mid f(x_i), \sigma^2 \sim \mathcal{N}(f(x_i), \sigma^2)$$

Gaussian noise: y = f(x) + ε, with ε ~ $\mathcal{N}$(0, σ²).

**HYPERPRIORS**

$$\alpha \sim p(\alpha), \ell \sim p(\ell), \sigma^2 \sim p(\sigma^2)$$

Fully Bayesian: priors on the kernel params and noise.

*Hierarchical:* $\alpha, \ell, \sigma^2 \to f \to y$

**POSTERIOR PREDICTIVE AT A NEW INPUT** · $f(x^*) \mid y \sim \mathcal{N}(\mu^*, v^*)$

**POSTERIOR MEAN**

$$\mu^* = m(x^*) + k^{*T}(K_\theta + \sigma^2 I)^{-1}(y - m)$$

*estimated function value at x**

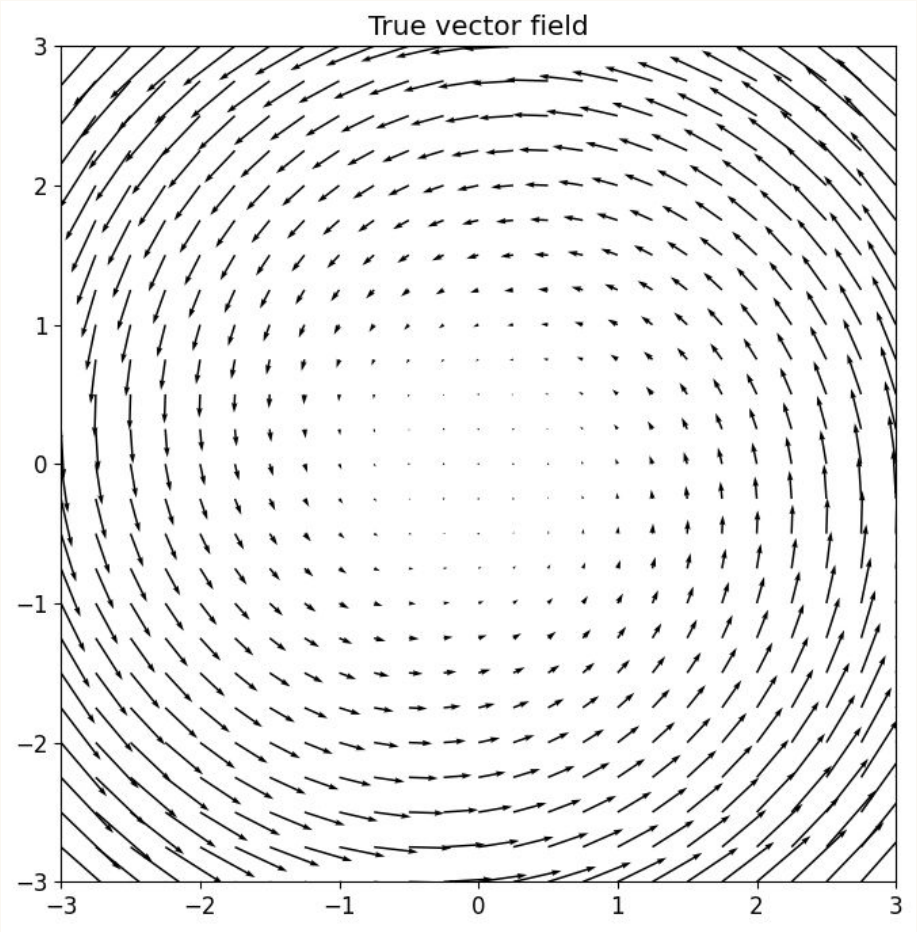
**POSTERIOR VARIANCE**

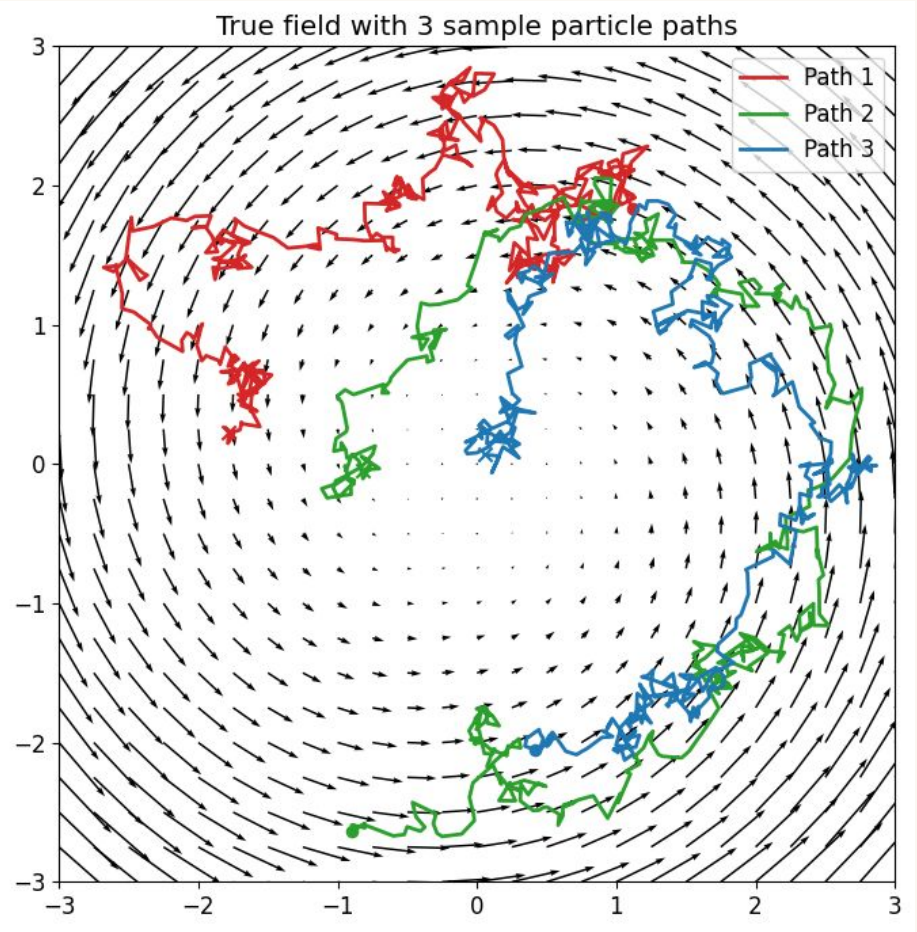
$$v^* = k(x^*, x^*) - k^{*T}(K_\theta + \sigma^2 I)^{-1}k^*$$

*principled uncertainty — shrinks near data*

# Toy Example: Recovering a Hidden Drift Field

Particles in $\mathbb{R}^2$ do a random walk (Gaussian noise) and are driven by unknown vector field $F$. We only observe the particle trajectories; we want to recover $F$.



# Toy Example: Recovering a Hidden Drift Field

Particles in $\mathbb{R}^2$ do a random walk (Gaussian noise) and are driven by unknown vector field $F$. We only observe the particle trajectories; we want to recover $F$.

$$X_{\square+1} = X_{\square} + F(X_{\square})\,\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_{\square} \quad \varepsilon_{\square} \sim N(0, I_2)$$

- $X_{\square}$ : current particle position
- $\sigma\sqrt{\Delta t}\,\varepsilon_{\square}$ : random perturbation over time $\Delta t$
- $F(X_{\square})\Delta t$ : deterministic drift from the hidden field over time $\Delta t$
- $\sigma > 0$ : sets the noise magnitude

**STEP 1 — Finite-difference velocity**

$$Y_{\square} := (X_{\square+1} - X_{\square}) / \Delta t$$

Rearranging the update yields $Y_{\square} = F(X_{\square}) + (\sigma/\sqrt{\Delta t})\,\varepsilon_{\square}$.

**STEP 2 — A regression problem emerges**

$$Y_i = F(x_i) + \eta_i$$

Each displacement is a **noisy local clue** about $F$. A GP prior on $F$ stitches them together.

# Fitting the GP to the Toy Example

The unknown object is the hidden vector field F; σ is treated as known.

**THE DATA** · Pairs $(x_i, Y_i)$ with $x_i, Y_i \in \mathbb{R}^2$ — a position and its noisy local velocity $Y_i = (X_{i+1} - X_i) / \Delta t$.

*Each short displacement is a noisy clue about $F$ at the current location.*

### LIKELIHOOD

$$Y_i \mid F(x_i), \sigma^2 \sim \mathcal{N}\left(F(x_i), (\sigma^2/\Delta t)\, I_2\right)$$

*Velocity centered on the true field; noise inflated by 1/Δt.*

### GP PRIOR ON THE FIELD

$$F_1 \sim \mathcal{GP}(0, k_\theta), \;\; F_2 \sim \mathcal{GP}(0, k_\theta)$$

*Two independent components, zero mean, shared RBF kernel $\theta = (\alpha, \ell)$.*

- *Base assumption is m(x)=0 everywhere (purely random walk)*

### HYPERPRIORS

$$\alpha \sim \text{Half-Normal}(2.5), \ell \sim \text{Half-Normal}(2)$$

*Weakly informative:*

- *$\ell$ : for a 6x6 observation window, allows for some local variation*
- *$\alpha$ : forces are same order of magnitude as particle stdv (0.5)*

### POSTERIOR TARGET

$$p(F_1, F_2, \alpha, \ell \mid Y_1, \ldots, Y_n)$$

*Update the prior over smooth vector fields using the observed local velocities.*

**IMPLEMENTATION** · Full Bayesian fit with **PyMC** (GP priors + MCMC)

# How the Posterior Sharpens with More Data

### 10 trajectories

- Posterior is still diffuse.
- Main directional signal detected, estimate of “center” is off.

### 20 trajectories

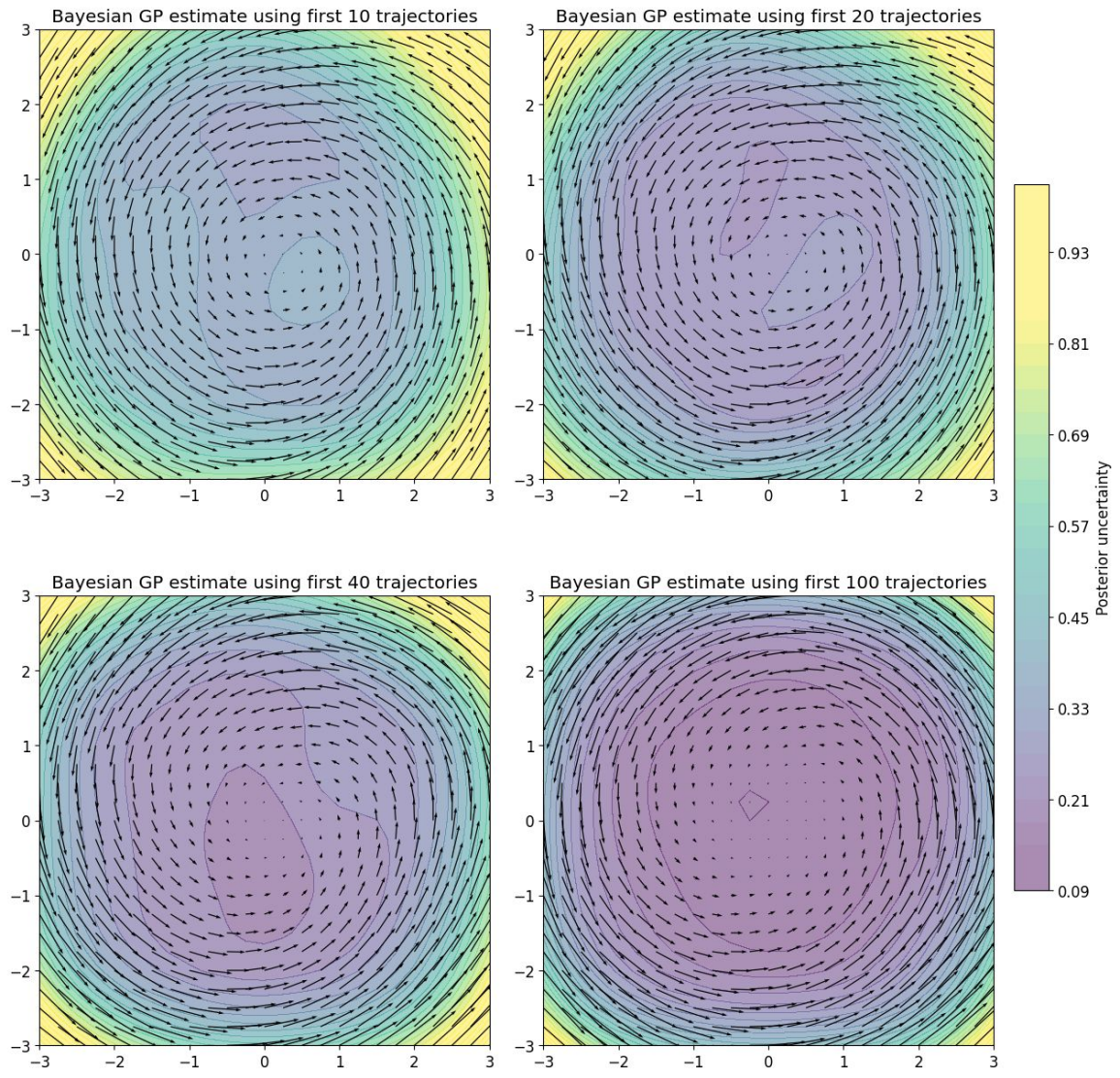
- Data are now strong enough to distinguish signal from noise in most of the interior.
- Posterior mean field is becoming stable.

### 40 trajectories

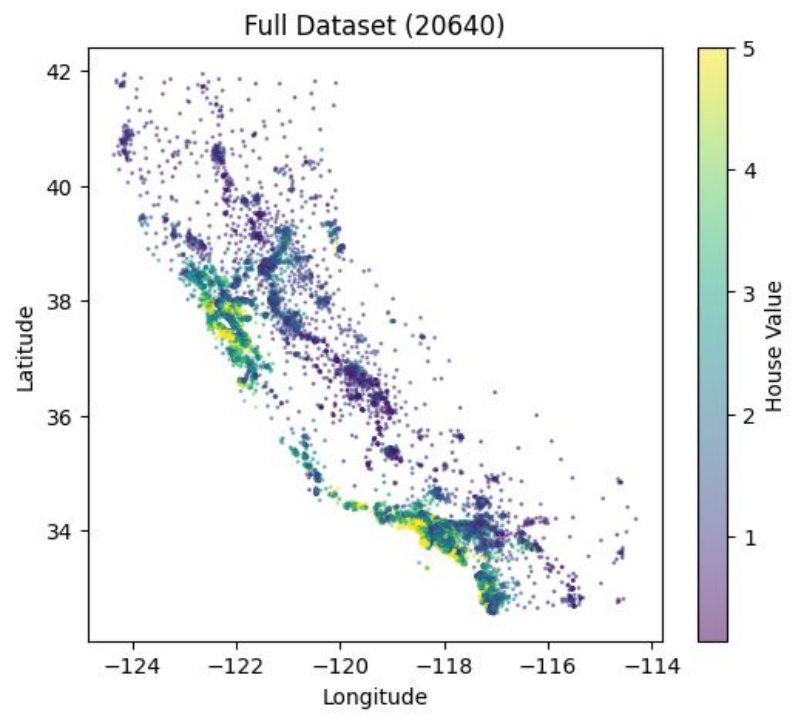
- Significant uncertainty remains around edges, reflecting lack of data.
- Global structure has converged.

### 100 trajectories

- Posterior has concentrated strongly around the true field.
- Posterior uncertainty reduced by an order of magnitude on majority of domain.



# Case Study: California Housing Dataset




**Full Dataset (N=20640)**

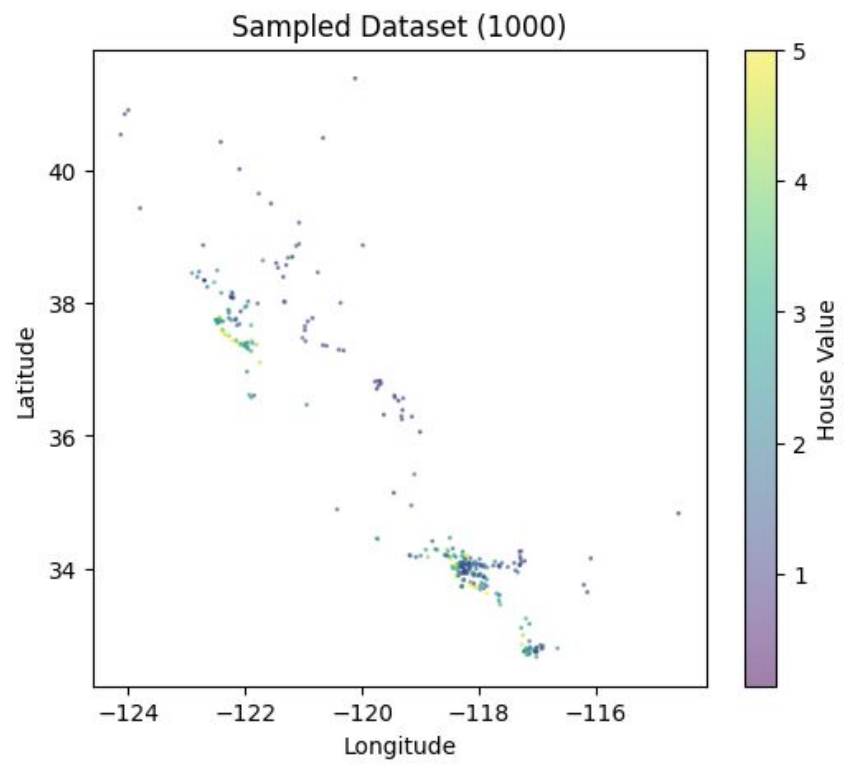
## Dataset Overview

Based on the 1990 U.S. Census. Each row is a "block group" (600–3,000 people); target is median house value.

| Column | Type | Description |
|---|---|---|
| `MedInc` | Float | Median income in the block group |
| `HouseAge` | Float | Median house age in the block group |
| `AveRooms` | Float | Average number of rooms per household |
| `AveBedrms` | Float | Average number of bedrooms per household |
| `Population` | Float | Total block group population |
| `AveOccup` | Float | Average number of household members |
| `Latitude` | Float | **Spatial input ($X_2$): block group latitude** |
| `Longitude` | Float | **Spatial input ($X_1$): block group longitude** |
| `MedHouseVal` | Float | **Target: median house value (×$100,000)** |

*Spatial inputs highlighted in blue; target variable highlighted in tan.*

# Standardization & Feature Selection




**Training Dataset (N=300)**

**Standardization:** We standardized the input features using z-scoring, important for Gaussian Process models since kernels measures distances between inputs.

**Feature Selection:** We have eight features available but used only longitude and latitude.

- **Inductive Bias:** What scale should relate longitude, latitude, house, age, and median income? Using the RBF kernel for all of these implies they are comparable to one another in a meaningful way, which they are not.
- **Curse of Dimensionality:** 8-dimensional data is sparse and would require a more flexible kernel.
- **Computation:** Costs scale $O(n^3)$ with the number of features, and the posterior geometry becomes complex, requiring longer chains.

# Specifying the Model

### PRIOR ON THE FUNCTION

$$f \sim \mathcal{GP}(0, k_\theta)$$

*Zero-mean GP with RBF (squared exponential) kernel.*

### LIKELIHOOD

$$y_i \mid f(x_i), \sigma^2 \sim \mathcal{N}(f(x_i), \sigma^2)$$

*Gaussian noise: $y = f(x) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.*

### HYPERPRIORS

$$\alpha \sim LogNormal(0, 0.5)$$
$$\rho \sim LogNormal(0, 0.5)$$
$$\sigma \sim LogNormal(-1, 0.5)$$

*Weakly informative priors on positive ($\alpha$, $\ell$, $\sigma$).*

*Hierarchical:* $\alpha, \ell, \sigma \to f \to y$

## Joint Distribution

Prior & Observations

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K(X,X) + \sigma^2 I & K(X, X_*) \\ K(X_*, X) & K(X_*, X_*) \end{bmatrix}\right)$$

### POSTERIOR PREDICTIVE AT A NEW INPUT · $f(x^*) \mid y \sim \mathcal{N}(\mu^*, v^*)$

### POSTERIOR MEAN

$$\mu^* = k^{*T}(K_\theta + \sigma^2 I)^{-1} y$$

### POSTERIOR VARIANCE

$$v^* = k(x^*, x^*) - k^{*T}(K_\theta + \sigma^2 I)^{-1} k^*$$

# Stan Code

## MODEL BLOCK

```
model {
  matrix[N_train, N_train] K;
  matrix[N_train, N_train] L_K;

  // Weakly informative priors for positive
  // GP hyperparameters
  alpha ~ lognormal(0, 0.5);
  rho   ~ lognormal(0, 0.5);
  sigma ~ lognormal(-1, 0.5);

  // Use robust gp_exp_quad_cov to avoid
  // 0 * Inf gradient NaNs
  K = gp_exp_quad_cov(X_train, alpha, rho);
  for (i in 1:N_train) {
    K[i, i] = K[i, i] + square(sigma) + jitter;
  }
  L_K = cholesky_decompose(K);
  y_train ~ multi_normal_cholesky(
    rep_vector(0, N_train), L_K);
}
```

## GENERATED QUANTITIES BLOCK

```
generated quantities {
  // Variable definitions redacted for brevity

  // Rebuild K using the built-in function
  K = gp_exp_quad_cov(X_train, alpha, rho);
  for (i in 1:N_train) {
    K[i, i] = K[i, i] + square(sigma) + jitter;
  }
  // Cross and test covariance
  K_s  = gp_exp_quad_cov(X_train, X_test, alpha, rho);
  K_ss = gp_exp_quad_cov(X_test, alpha, rho);
  for (i in 1:N_test) {
    K_ss[i, i] = K_ss[i, i] + jitter;
  }
  // Posterior predictive mean and covariance
  alpha_vec   = mdivide_left_spd(K, y_train);
  f_test_mean = K_s' * alpha_vec;
  K_pred      = K_ss - K_s' * mdivide_left_spd(K, K_s);
  // Add observation noise + jitter to stabilize
  for (i in 1:N_test) {
    K_pred[i, i] = K_pred[i, i] + square(sigma) + jitter;
  }
  // Draw posterior predictive samples
  y_test_rep = multi_normal_rng(f_test_mean, K_pred);
}
```

# Numeric Tricks for GP Stability

## Ensuring Numerical Stability in GP Models

### 01. Safe Boundaries & Priors

Applied `<lower=1e-6>` constraints and Log-normal priors to prevent parameters from collapsing to zero.

### 02. Diagonal Jitter

Added `(1e-5)` to the covariance matrix to guarantee positive-definite.

### 03. Optimized Built-in Functions

Utilized `gp_exp_quad_cov` to avoid 0 * Inf gradient crashes during HMC warmup exploration.

### 04. Cholesky & SPD Solvers

Avoided direct matrix inversions with robust Cholesky decompositions and `mdivide_left_spd` solvers.

# Diagnostics

**Summary Statistics (alpha, rho, sigma)**

| | Mean | StdDev | R_hat | ESS_bulk | ESS_tail |
|---|---|---|---|---|---|
| alpha | 0.918281 | 0.233114 | 1.00542 | 1158.900 | 663.511 |
| rho | 0.389081 | 0.290146 | 1.00602 | 750.555 | 801.452 |
| sigma | 0.772572 | 0.059097 | 1.00153 | 1049.910 | 922.148 |

**Quantile & Convergence Metrics**

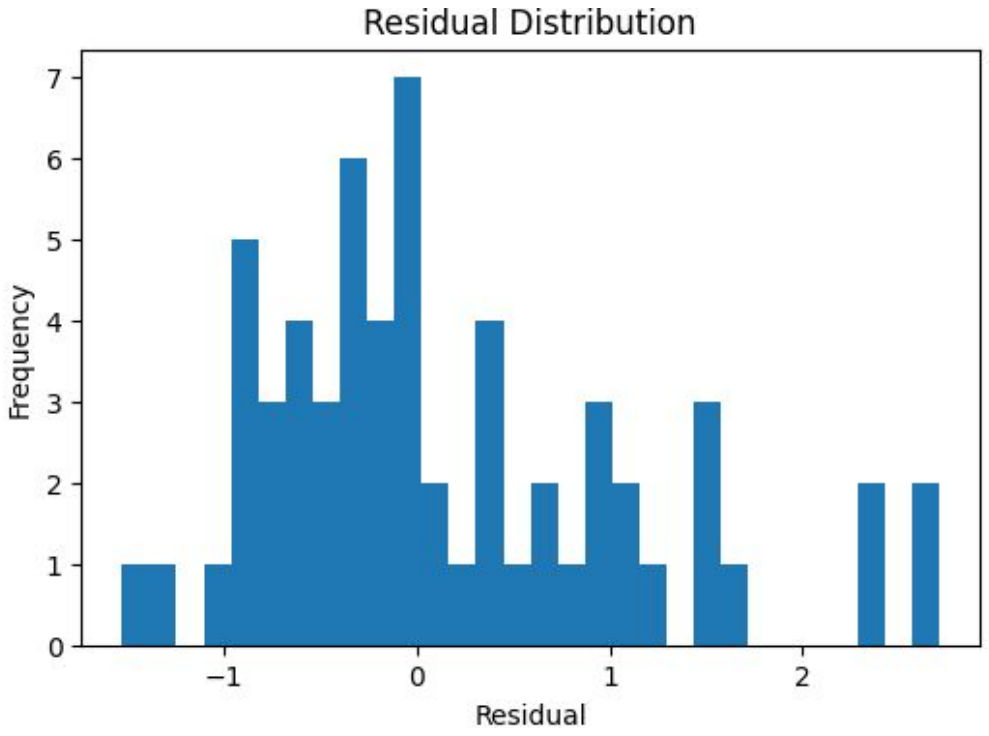
| | Mean | StdDev | 5% | 50% | 95% | R_hat |
|---|---|---|---|---|---|---|
| alpha | 0.918281 | 0.233114 | 0.651976 | 0.873101 | 1.32777 | 1.00542 |
| rho | 0.389081 | 0.290146 | 0.096951 | 0.299800 | 1.01812 | 1.00602 |
| sigma | 0.772572 | 0.059097 | 0.669029 | 0.774810 | 0.86445 | 1.00153 |



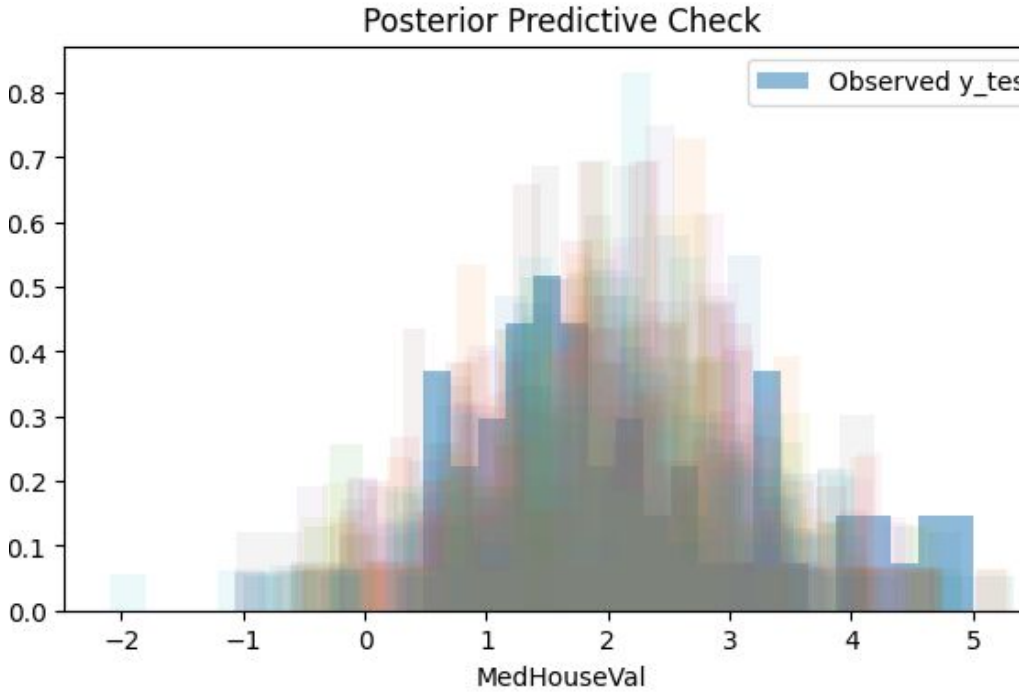
*Gaussian Processes*

# Diagnostics

## Residual Distribution



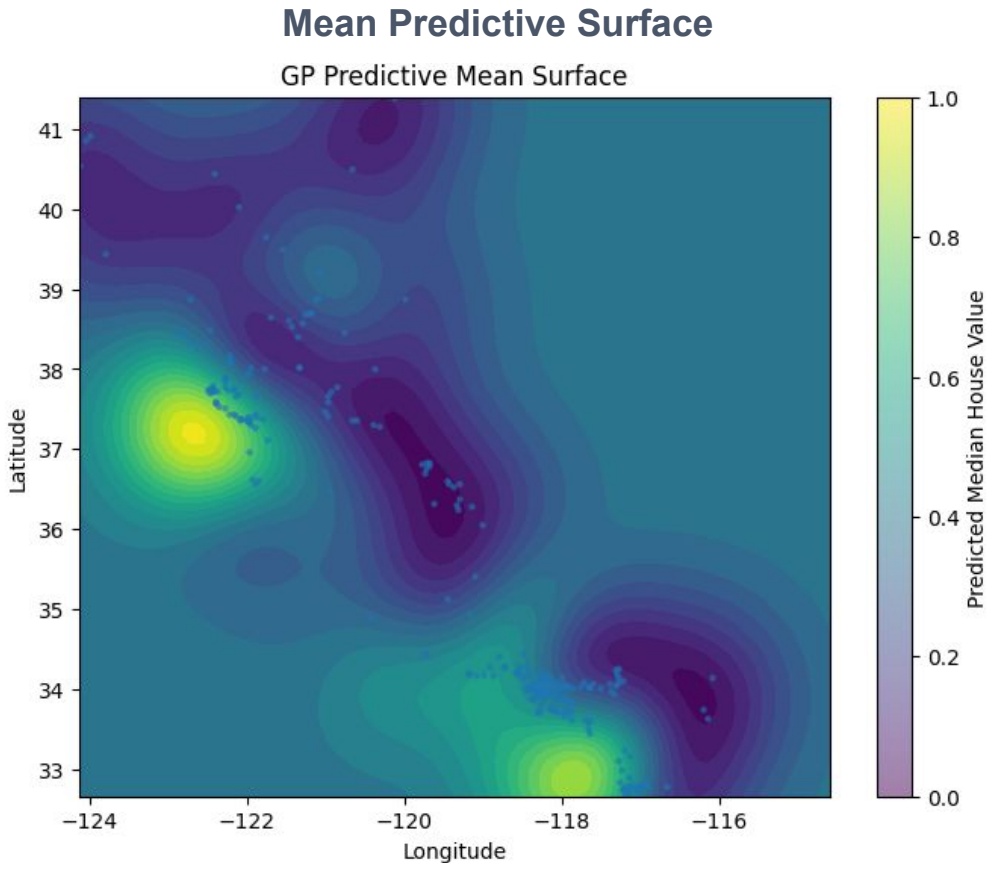

## Posterior Predictive Check

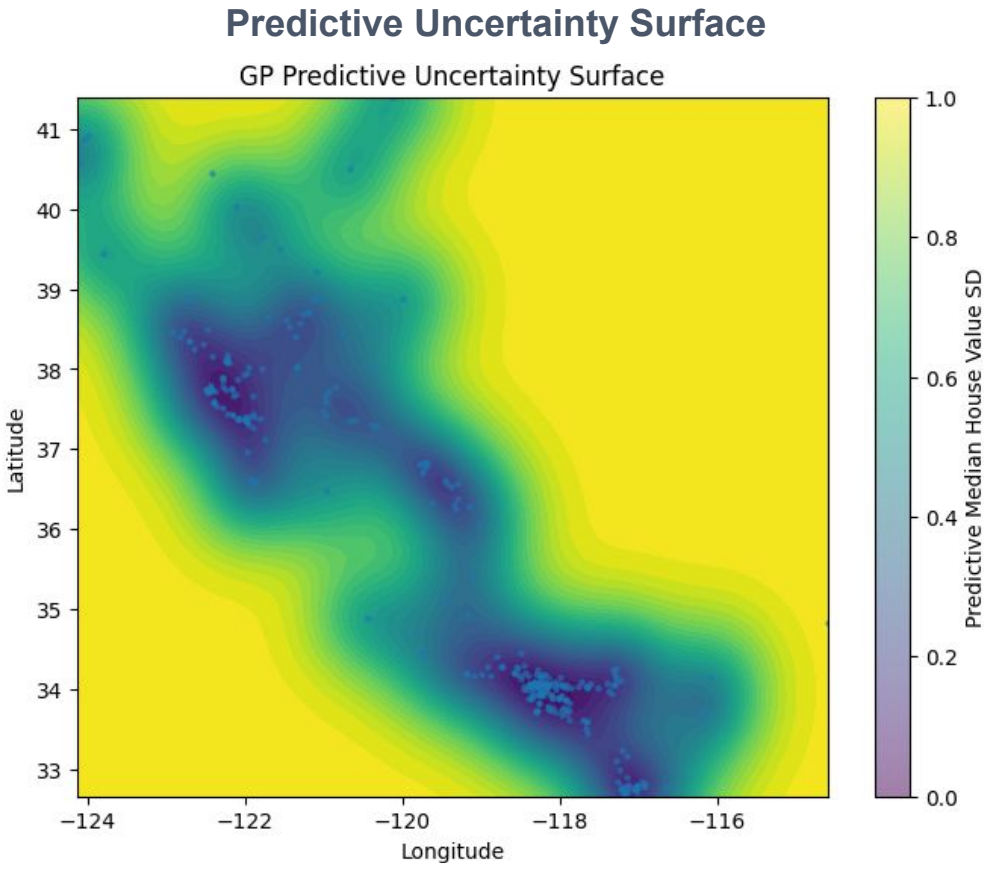





Gaussian Processes

# Reconstructed GP: Posterior Heatmap






*Gaussian Processes*

# Areas for Improving the Model

## Oversmoothing from the RBF kernel

The squared-exponential (RBF) kernel assumes a very smooth latent function modulated only by $\rho$, washing out sharp local changes and underestimating extreme prices.

$$k(x, x') = \alpha^2 \exp(-\|x - x'\|^2 / 2\rho^2)$$

## Composite spatial + demographic kernels

Split x = (s, z) with s = (Lat, Lon) and combine two kernels — capturing smooth coordinates alongside nonlinear demographic features.

$$k(x, x') = k_{spatial}(s, s') + k_{demographic}(z, z')$$

## A richer kernel with more features

Augment the input with all predictors (Lat, Lon, MedInc, HouseAge, AveRooms, AveBedrms, Pop, AveOccup) and use an Auto-Relevance Detector (ARD) kernel with dimension-specific length scales ρ□.

$$k(x, x') = \alpha^2 \exp(-\tfrac{1}{2} \Sigma_{\square} (x_{\square} - x'_{\square})^2 / \rho_{\square}^2)$$

## Inclusion of linear structure

Combine a linear mean with GP so large-scale linear effects (e.g., median income) are modeled directly, while the GP captures nonlinear structure.

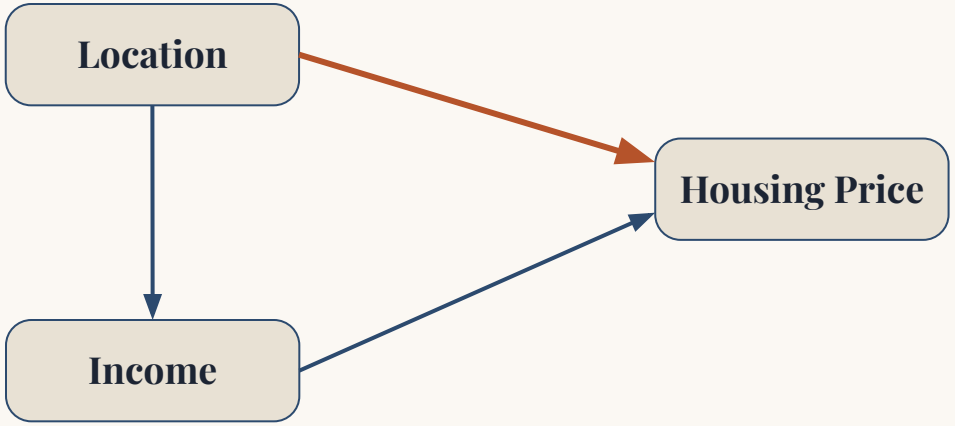
$$y_i = \beta_0 + x_i^T \beta + f(x_i) + \varepsilon_i$$

# Augmenting the Model with Income

$$k(x, x') = k_{spatial}(s, s') + k_{income}(z, z')$$

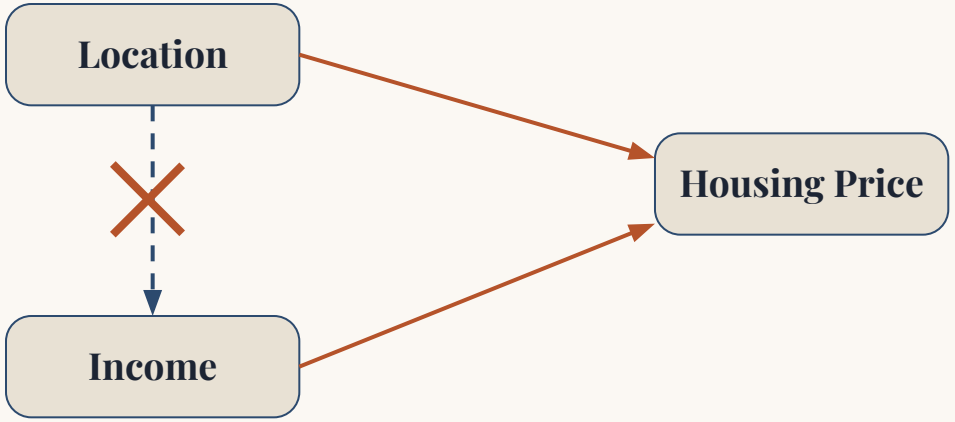
*An additive income kernel helps eliminate* ***omitted variable bias***

## Geography-only model




Income affects housing prices and varies geographically. By only modeling Location ↠ Housing Price, we force our spatial kernel to implicitly capture the location-income covariance as well.

## With income variable added




By introducing the income variable, the spatial kernel no longer attempts to explain the indirect effect of Location ↠ Income ↠ Housing Price.

# Inside the Augmented Model

## Spatial kernel

$$k_{spatial}(s, s') = \alpha_{loc}{}^2 \exp(-\|s - s'\|^2 / 2\rho_{loc}{}^2)$$

- $\alpha_{loc}$ controls the size of the spatial variation
- $\rho_{loc}$ is the spatial lengthscale
- Nearby locations are more strongly correlated

## Income kernel

$$k_{income}(z, z') = \alpha_{inc}{}^2 zz'$$

A linear kernel — the income effect is linear, but its slope is uncertain and learned from the data.

- $\alpha_{inc}$ controls the strength of the income signal

## Full covariance structure

$$K = K_{loc} + K_{inc} + \sigma^2 I$$

$$y_{train} \sim N(0, K)$$

Price variation = smooth spatial effect + linear income effect + residual noise.

## Equivalent latent-function decomposition

$$f(s, z) = f_{loc}(s) + f_{inc}(z)$$

$$f_{loc} \sim \mathcal{GP}(0, k_{spatial}),\ f_{inc} \sim \mathcal{GP}(0, k_{income})$$

The total GP is additive — one component for space, one for income.

# Relationship with the Mean Function

## From Bayesian linear effect to linear kernel

Suppose we had instead modeled the income contribution as a mean function with normal $\beta$ hyperprior:

$$m(z) = \beta z, \quad \beta \sim N(0, \alpha_{inc}{}^2)$$

For two incomes $z$ and $z'$, what is the induced covariance?

$$Cov(\beta z, \beta z') = zz'\mathbb{E}[\beta^2] = zz' \, Var(\beta) = \boxed{\alpha_{inc}{}^2 \, zz'}$$

## The equivalence, in plain terms

- If the data are centered, a linear mean function with a Gaussian prior on the coefficient is mathematically equivalent — after marginalizing over the coefficient — to a zero-mean GP with a linear kernel.
- So the linear income kernel is the GP representation of Bayesian linear regression on income.

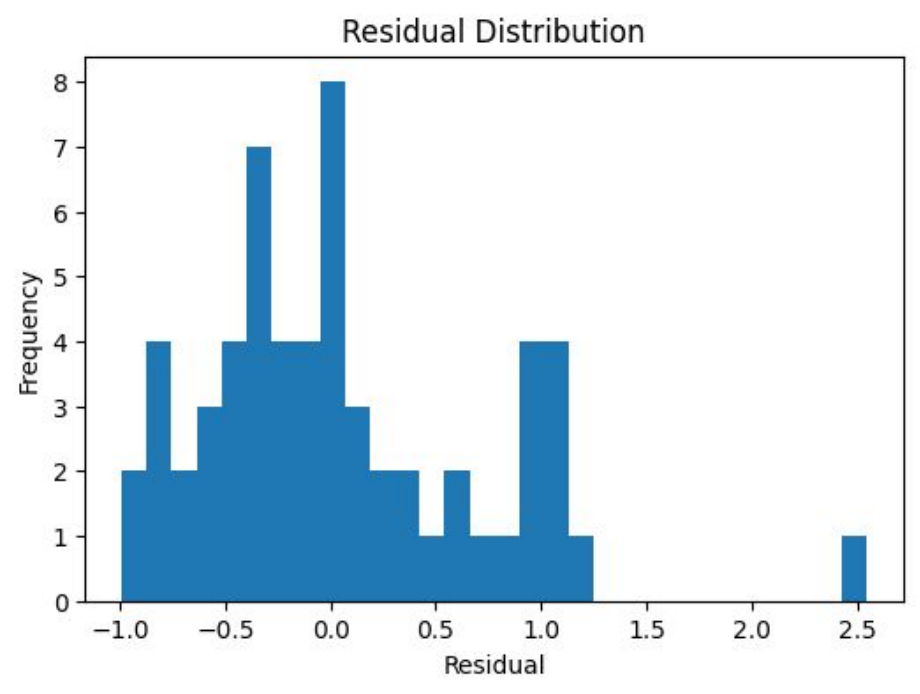
## Why this matters

Splitting the kernel additively lets us separate two kinds of structure:

- Nonlinear smooth spatial dependence (RBF on location)
- Linear dependence on income (Bayesian linear effect)
- This reduces the risk that the spatial kernel incorrectly absorbs income-driven covariance

# Diagnostics — Augmented Model

## Residual Distribution

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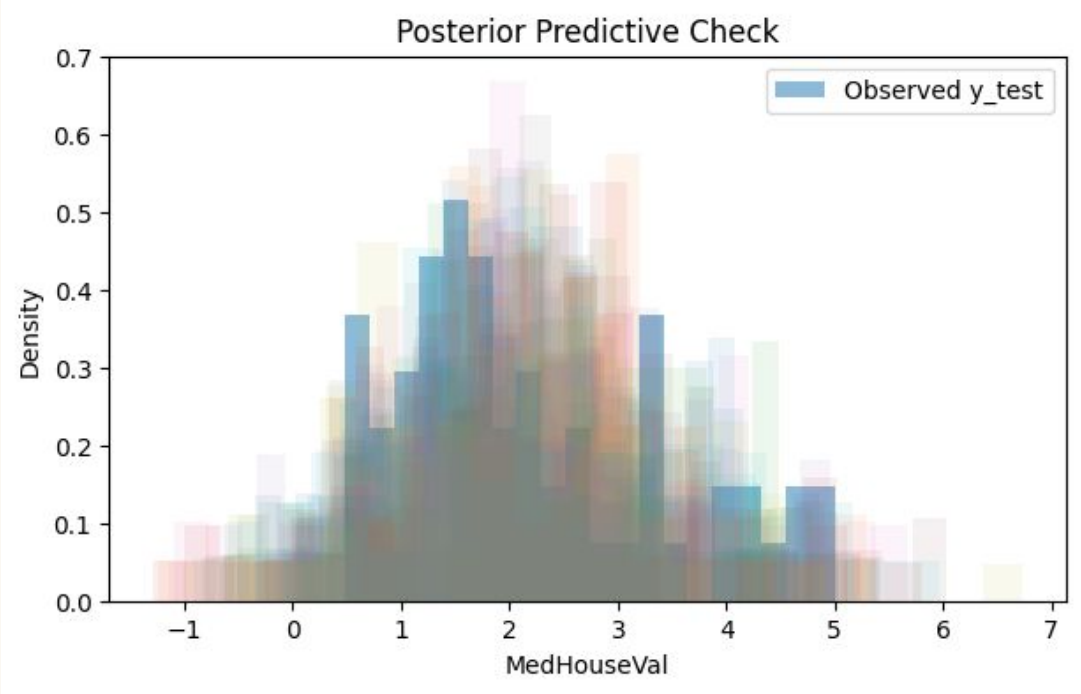
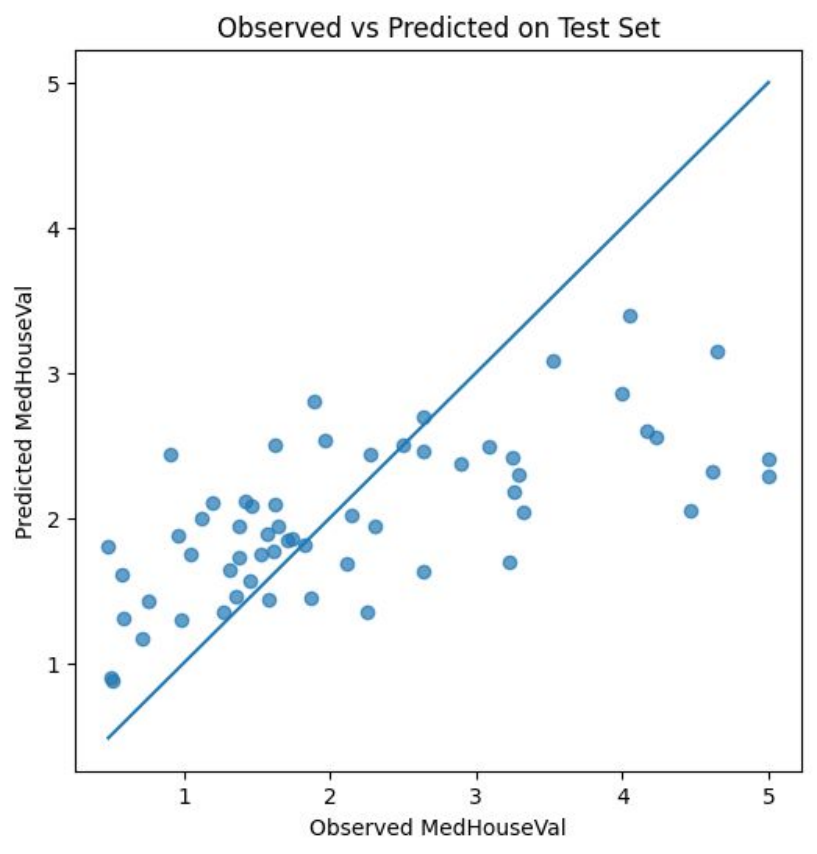


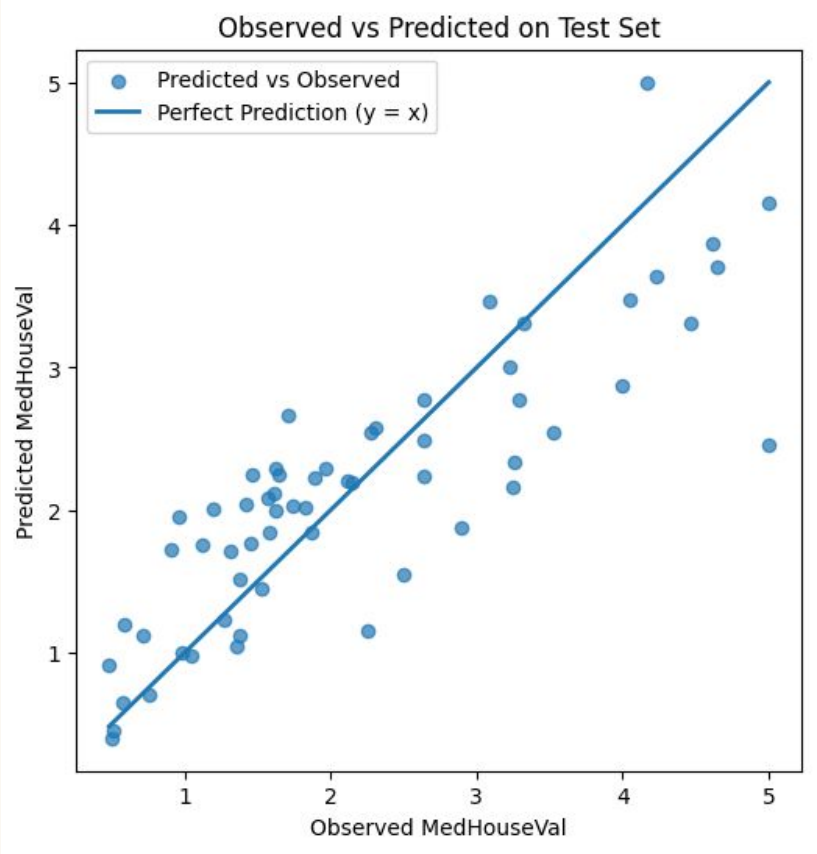
# Out-of-Sample Performance

## Spatial-only model




*Predictions vs. actuals on held-out test set*

## Spatial + Income model




*Predictions vs. actuals on held-out test set*

# Thank You

*Questions?*